数 学 作 业 纸

科目________

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6.2b $M_1 = -30kN \times 2 = -60kN\cdot m$

$M_2 = +40kN\cdot m.$

$M = -60kN\cdot m + 40kN\cdot m = -20kN\cdot m.$

$I_z = \frac{bh^3}{12} = \frac{80\times120^3}{12} = 11.52\times10^6\ mm^4$

$W_z = \frac{bh^2}{6} = \frac{80\times120^2}{6} = 192\times10^3\ mm^3$

$\sigma_{max} = \frac{|M|}{W_z} = \frac{20\times10^6}{192\times10^3} \approx 104.17MPa.$

$y_a = 30mm.$ $\sigma_a = \frac{|M|y_a}{I_z} = \frac{20\times10^6\times30}{11.52\times10^6} \approx 52.08MPa.$

6.2a. $M_{max} = \frac{ql^2}{8} = \frac{10\times2^2}{8} = 5kN\cdot m.$

$I_z = \frac{\pi D^4}{64} = \frac{\pi\times80^4}{64} \approx 2.01\times10^6\ mm^4.$

$W_z = \frac{\pi D^3}{32} \approx 50.27\times10^3\ mm^3$

$\sigma_a = \frac{My_a}{I_z} = \frac{5\times10^6\times30}{2.01\times10^6} \approx 74.6MPa$(压应力.)

$\sigma_{max} = \frac{M}{W_z} = \frac{5\times10^6}{50.27\times10^3} \approx 99.5MPa.$

6.5 $\Sigma M_B = 0$

$20kN\times3m + 40kN\times1m - R_A\times2m = 0 \Rightarrow 2R_A = 100 \Rightarrow R_A = 50kN$(向上)

$\Sigma M_A = 0.$

$20kN\times1m - 40kN\times1m + R_B\times2m = 0 \Rightarrow 2R_B = 20 \Rightarrow R_B = 10kN$(向上).

$M_A = -20kN\times1m = -20kN\cdot m$

$M_{40} = R_B\times1m = 10kN\times1m = 10kN\cdot m$

$|M|_{max} = 20kN\cdot m = 20\times10^3 N\cdot m.$

$\sigma_{max} = \frac{|M|_{max}}{W_z} = \frac{20\times10^3}{141\times10^{-6}} \approx 141.84\times10^6 Pa = 141.84MPa.$

$\sigma_{max} = 141.84MPa \leq [\sigma] = 160MPa.$ 正应力强度满足要求.

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6.17. $R_B \times 2m - 40kN \times 2.5m = 0 \Rightarrow R_B = 50kN$

$|M|_{max} = 40kN \times 0.5m = 20kN \cdot m = 20 \times 10^3 N \cdot m$

$W_z \geq \dfrac{|M|_{max}}{[\sigma]} = \dfrac{20 \times 10^3}{160 \times 10^6} = 125 \times 10^{-6} m^3 = 125 cm^3$.

16号工字钢 $W_z = 141 cm^3$. 选取16号工字钢.